

Property Rental Management System

A relational database application built on Oracle APEX & PL/SQL

Created By: Lavanya Kapoor

PROJECT DESCRIPTION

THE PROBLEM

- ✗ Paper lease agreements — slow, lossy
- ✗ Excel rent tracking — no overdue alerts
- ✗ Email maintenance — requests go missing
- ✗ No landlord portfolio visibility
- ✗ No unified search across records

OUR SOLUTION

- ✓ Centralized Oracle APEX database app
- ✓ 8 relational tables with FK constraints
- ✓ Automated payment & lease tracking
- ✓ Role-aware landlord portfolio dashboard
- ✓ Universal search across all data

WHAT FUNCTIONS DOES THE SYSTEM PROVIDE?



Property Management

Add, edit & track all rental properties with occupancy status, rent, type & landlord link



Tenant Registration

Register tenants with ID verification, credit score, emergency contact & move-in date



Lease Management

Create leases linking tenant + property, auto-set status to Active, enforce date rules



Payment Tracking

Record rent payments, auto-detect Paid/Partial/Overdue, maintain full audit trail



Maintenance Requests

Submit & assign repair requests by priority - Emergency flags property as Under Maintenance



Universal Search Engine

Search Properties, Tenants & Leases simultaneously from one screen using one query



Reports & Dashboard

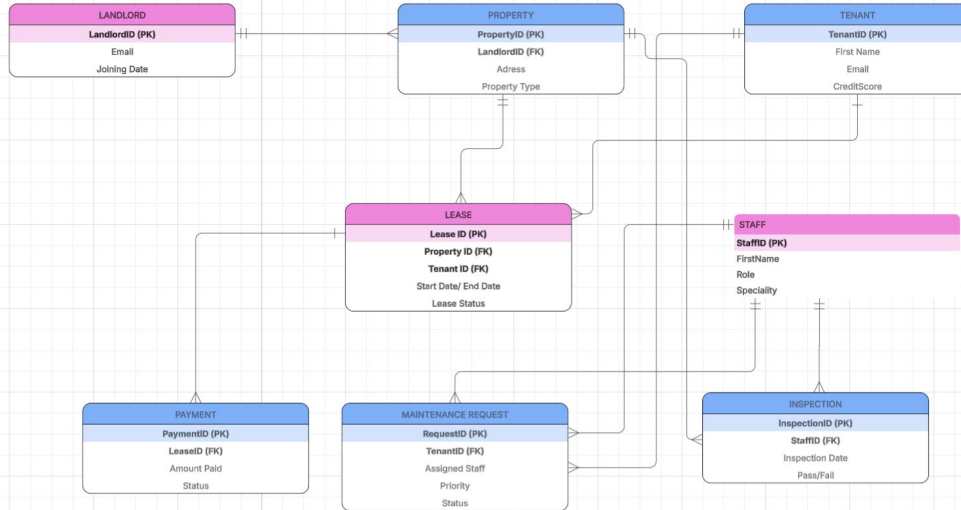
9 live reports: overdue payments, expiring leases, landlord portfolio, inspections & more



Inspection Logging

Log Move-In/Move-Out/Routine inspections with Pass/Fail and next due date

ER DIAGRAM & ASSUMPTIONS



KEY ASSUMPTIONS

→ One landlord → many properties (1:M)

→ Lease resolves Tenant ↔ Property M:M

→ Payment depends on Lease (weak entity)

→ EndDate must be > StartDate

→ CreditScore valid: 300–850

→ Emergency request → flags property

→ Cascade delete: Lease → Payments (If a Lease is ever deleted, automatically clean up all its payments too.)

→ One property = one active lease at a time

QUERIES, FORMS, REPORTS & SEARCH ENGINE

QUERIES

CREATE TABLE

→ Build all 8 tables with columns & rules

INSERT INTO

→ Add landlords, tenants, leases, payments

SELECT + JOIN

→ View data across linked tables together

COUNT / GROUP

→ Count properties, payments, requests

WHERE / LIKE

→ Filter by status, date, name, city

REPORTS — What You Can See

Property Listing • Tenant Directory • Active Leases
Payment Tracking • Overdue Payments • Open Maintenance
Landlord Portfolio • Expiring Leases • Inspection History

FORMS - Add Data Easily

✓ Property Form — add a new rental unit

✓ Tenant Form — register a new tenant

✓ Lease Form — create a lease agreement

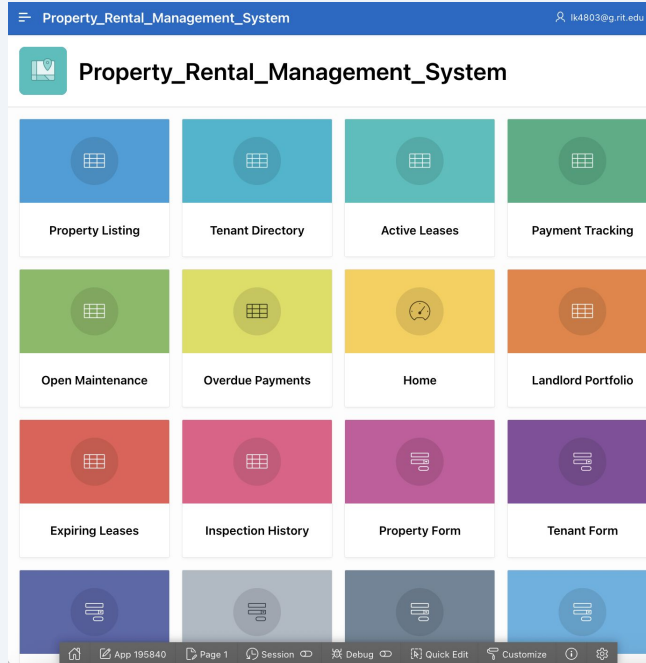
✓ Payment Form — record a rent payment

✓ Maintenance Form — log a repair request

SEARCH ENGINE — One Box, Everything

Type anything — name, city, email, status
Searches Properties + Tenants + Leases at once

APPLICATION : HOME PAGE & DASHBOARD



PRMS Dashboard

Metric	Value ↓
Total Properties	17
Total Tenants	12
Active Leases	11
Occupied Properties	8
Open Maintenance	8
Total Staff	7
Available Properties	6
Overdue Payments	4

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All 16 pages accessible from home • Live dashboard shows: Properties, Leases, Tenants, Overdue, Maintenance counts

FORMS : Data Entry with Smart Dropdowns

Property_Rental_Management_System

Row created.

Property Form

Property

Landlordid

Robert

Address

55 Oak Street

City

Albany

State

NY

Zip

12201

Propertytype

Apartment

Bedrooms

2

Bathrooms

1

Soft

800

Rentamount

1100

Dep

220

App 195840

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Session

Debug

Quick Edit

Customize

Property_Rental_Management_System

Row created.

Payment Form

Payment

Leaseid

Active

Tenantid

Emma

Amountpaid

1100

Duedate

5/1/2026

Paiddat

5/1/2026

Paymentmethod

Bank Transfer

Referenceno

TXN-99999

Status

Paid

Cancel

Delete

Apply Changes

REPORTS

Property_Rental_Management_System

448023@pg.it.edu

Home

Property Listing

Tenant Directory

Active Leases

Payment Tracking

Open Maintenance

Overdue Payments

Home

Landlord Portfolio

Expiring Leases

Inspection History

Property Form

Tenant Form

Lease Form

Payment Form

Maintenance Form

Inspection Form

Search

Property Listing

Q

Go

Actions

Reset

LandlordId	Address	City	State	Zip	PropertyType	Bedrooms	Bathrooms	Soft	Rentamount	Depositamount	Status	Yearbuilt
James	7 Birch Blvd	Syracuse	NY	13201	Apartment	1.0	1.0	600	900	1,800	Under Maintenance	2,016
James	30 Onondaga Ave	Syracuse	NY	13204	House	3.0	2.0	1,350	1,650	3,300	Occupied	1,995
James	62 Salt Springs	Syracuse	NY	13208	Condo	2.0	1.0	900	1,100	2,200	Available	2,008
Linda	99 Willow Way	Buffalo	NY	14202	House	4.0	2.5	2,000	2,200	4,400	Available	2,000
Linda	55 Niagara Blvd	Buffalo	NY	14207	Apartment	1.0	1.0	620	950	1,900	Occupied	2,012
Linda	12 Cedar Ln	Buffalo	NY	14201	Condo	2.0	1.5	1,100	1,400	2,800	Occupied	2,010
Patricia	15 Park Place	Albany	NY	12205	Studio	1.0	1.0	480	800	1,600	Available	2,019
Patricia	9 Capitol View	Albany	NY	12201	Apartment	2.0	1.0	780	1,150	2,300	Occupied	2,014
Patricia	47 Hudson St	Albany	NY	12202	House	4.0	3.0	1,800	2,100	4,200	Occupied	2,003
Robert	55 Oak Street	Albany	NY	12201	Apartment	2.0	1.0	800	1,100	2,200	Available	2,012
Robert	100 Test Street	Rochester	NY	14607	Apartment	2.0	1.0	900	1,100	2,200	Available	2,010
Robert	200 River Rd	Rochester	NY	14613	Condo	2.0	2.0	1,050	1,350	2,700	Inactive	2,001
Robert	14 Spruce Ct	Rochester	NY	14612	Studio	1.0	1.0	500	850	1,700	Occupied	2,018
Robert	45 Maple St	Rochester	NY	14607	Apartment	2.0	1.0	850	1,200	2,400	Occupied	2,005
Robert	88 Elm Dr	Rochester	NY	14610	House	3.0	2.0	1,400	1,800	3,600	Available	1,986
Thomas	22 Troy Hill Rd	Troy	NY	12180	House	3.0	2.0	1,250	1,550	3,100	Occupied	1,999
Thomas	88 Federal St	Troy	NY	12182	Apartment	2.0	1.0	720	1,050	2,100	Under Maintenance	2,007

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Property_Rental_Management_System

448023@pg.it.edu

Home

Property Listing

Tenant Directory

Active Leases

Payment Tracking

Open Maintenance

Overdue Payments

Home

Landlord Portfolio

Expiring Leases

Inspection History

Property Form

Tenant Form

Lease Form

Payment Form

Tenant Directory

Q

Go

Actions

Reset

Phone	Email
585-555-0201	m.gonzalez@email.com
585-555-0202	j.brown@email.com
585-555-0203	p.patel@email.com
585-555-0204	k.lee@email.com
585-555-0205	s.thompson@email.com
585-555-0206	d.martinez@email.com
585-555-0207	a.wilson@email.com
585-555-0208	m.taylor@email.com
585-555-0209	j.davis@email.com
585-555-0210	r.garcia@email.com
585-555-0211	j.miller@email.com
585-555-0212	e.scott@gmail.com

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UNIVERSAL SEARCH ENGINE

Property_Rental_Management_System

lk4803@g.rit.edu

Search

Search

j.brown

Search

Search Results

Recordtype	Recordid	Description	Statusfield	Extrainfo
Tenant	2	James Brown	Verified	j.brown@email.com

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Search

Search

Active

Search

Search Results

Recordtype	Recordid	Description	Statusfield	Extrainfo
Lease	10	Lease #10 Property 12	Active	Rent: \$2,100.00
Lease	12	Lease #12 Property 14	Active	Rent: \$1,550.00
Lease	13	Lease #13 Property 12	Active	Rent: \$2,100.00
Lease	2	Lease #2 Property 1	Active	Rent: \$1,250.00
Lease	21	Lease #21 Property 5	Active	Rent: \$1,800.00
Lease	4	Lease #4 Property 3	Active	Rent: \$850.00
Lease	6	Lease #6 Property 5	Active	Rent: \$1,450.00
Lease	7	Lease #7 Property 7	Active	Rent: \$950.00
Lease	8	Lease #8 Property 9	Active	Rent: \$1,650.00
Lease	9	Lease #9 Property 11	Active	Rent: \$1,150.00

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HOW IT WORKS

One search box → UNION ALL query → searches Properties + Tenants + Leases simultaneously → wildcard LIKE matching → Result at Oracle APEX

LIVE DEMO

Switch to Oracle APEX — App 195840

- | | | |
|----|------------------|---|
| 01 | Home Page | → Show the navigation cards — all modules in one view |
| 02 | Dashboard | → Scroll down — show live metrics: 17 props, 11 leases, 4 overdue |
| 03 | Property Listing | → Click Property Listing — show all 17 properties, landlord names, status |
| 04 | Overdue Payments | → Click Overdue Payments — 4 tenants with days overdue + contact info |
| 05 | Maintenance Form | → Click Maintenance Form — fill in a new request, show dropdown, submit |
| 06 | Search Engine | → Go to Search — type 'Buffalo' → properties; type 'Priya' → tenant |
| 07 | Expiring Leases | → Click Expiring Leases — show 5 leases expiring within 60 days |

LESSONS LEARNED



Design the ERD first - before any SQL

Planning all 8 entities and relationships upfront prevented major rework. Every table we created matched the ERD exactly.



SQL Scripts ≠ SQL Commands in Oracle APEX

SQL Commands runs one statement at a time. Multi-block PL/SQL with CREATE PROCEDURE needs SQL Scripts with '/' terminators — we learned this the hard way after multiple PLS-00103 errors.



Views make APEX report pages instant

Pre-joining 3-4 tables into a view meant APEX report pages needed zero extra SQL — just point at the view. This saved hours of debugging.



Test data quality matters more than quantity

Having mixed statuses (Paid/Overdue/Partial, Active/Expired, Open/Resolved) made every report meaningful. Uniform data produces boring reports.

PROJECT LIMITATIONS & FUTURE WORK



Current Limitations

✗ No role-based login — anyone can see everything

✗ No automated email alerts for overdue payments

✗ No PDF upload for actual lease documents

✗ No concurrent access control for multiple users



Future Work

✓ Role-based login: landlord vs staff vs admin view

✓ Automated overdue payment email notifications

✓ Tenant self-service portal for maintenance

✓ Integration with payment gateways (Stripe etc.)

Thank You!

Property Rental Management System

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Questions?

Oracle APEX App 195840 • 8 Tables • 7 Procedures
9 Reports • 1 Search Engine • 17 Properties • 12 Tenants